

AMTI (NMTC) - 2015

RAMANUJAN CONTEST - FINAL - INTER LEVEL

1. a) If x and y are positive reals such that $x^{2014} + y^{2014} = 1$ prove that

$$\left(\sum_{k=1}^{1007} \frac{1+x^{2k}}{1+x^{4k}} \right) \left(\sum_{k=1}^{1007} \frac{1+y^{2k}}{1+y^{4k}} \right) < \frac{1}{(1-x)(1-y)}.$$

- b) The angles of a triangle are in arithmetic progression. The altitudes of this triangle are also in arithmetic progression. Show that the triangle is equilateral.
2. ABC is an acute angled triangle in which the three sides are unequal. L, M, N are respectively the midpoints of the sides BC, CA and AB. The perpendicular bisectors of the sides AB and AC intersect AL at D and E respectively. BD and CE cut at F inside the triangle. Show that A, M, F, N are concyclic.
3. S is a finite set of positive integers such that for any two distinct numbers of S, a and b , the number $\frac{a+b}{\gcd(a,b)}$ is also a member of S. Determine all such sets S.

4. a) Find all real number triples (x, y, z) which satisfy

$$\begin{aligned} 3(x^2 + y^2 + z^2) &= 1 \\ x^2 y^2 + y^2 z^2 + z^2 x^2 &= xyz(x + y + z)^3 \end{aligned}$$

- b) Show that all the integers 1 to 16 can be written in a line, but not in a circle, so that the sum of any two adjacent numbers is a perfect square.
5. A certain number is the product of three different prime factors, the sum of whose squares is 2331. There are 7560 numbers (including unity) which are less than the number and prime to it. The sum of all its divisors (including unity and the number) is 10560. Find the number.
6. A rectangular parallelepiped is given, such that its intersection with a plane is a regular hexagon. Prove that the rectangular parallelepiped is a cube.
7. A triangle ABC is given. The midpoints of the sides AC and AB are B_1 and C_1 respectively. The incenter of the $\triangle ABC$ is I. The lines $B_1 I$ and $C_1 I$ meet the sides AB, AC at C_2 and B_2 respectively. If the areas of $\triangle ABC$ and $\triangle AB_2 C_2$ are equal, find the measure of the angle BAC.
8. If $(a+b)$ and $(a-b)$ are relatively prime integers for some natural numbers a, b . Find the greatest common divisor of $2a + (1-2a)(a^2 - b^2)$ and $2a(a^2 + 2a - b^2)(a^2 - b^2)$.
